

UART Design and Verification Project

Version 1 – Engineering Design Document

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Tools Used

- Verilog HDL
 - VS Code
- Siemens QuestaSim 2025
 - EDA Playground

Status: Version 1 Complete

Revision History

Version	Date	Author	Description
1.0	June 2026	Ashritha Vasthala	Initial RTL implementation

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1. Project Overview

Universal Asynchronous Receiver Transmitter (UART) is a widely used asynchronous serial communication protocol that enables reliable data transfer between digital systems without a shared clock. Communication is synchronized using a predefined baud rate and organized into frames containing a start bit, eight data bits and a stop bit in this implementation.

This project implements a modular UART subsystem using synthesizable Verilog HDL. Version 1 focuses on RTL implementation and functional verification. The design has been organized into independent modules to improve readability, maintainability and future scalability.

2. Project Objectives

- Develop synthesizable UART RTL using Verilog HDL.
- Implement a parameterized baud generator.
- Design UART transmitter and receiver modules.
- Integrate all RTL modules into a top-level design.
- Verify functionality using a Verilog testbench.
- Validate operation through simulation waveform analysis.

3. Scope of Version 1

Version 1 includes only the completed work:

- baud_generator.v
- uart_tx.v
- uart_rx.v
- uart_top.v
- uart_top_tb.v
- Integrated functional simulation and waveform verification.

Advanced features such as parity, FIFO buffering, assertions, functional coverage and UVM verification are outside the scope of Version 1.

4. Design Specifications

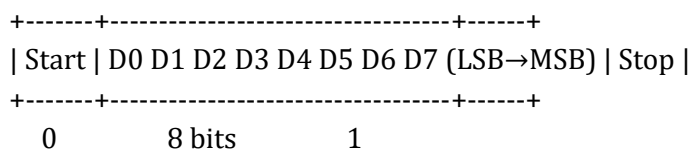
Parameter	Specification
Clock Frequency	50 MHz
Baud Rate	115200 bps
Data Width	8 bits
Parity	Not Implemented
Stop Bits	1
RTL Language	Verilog HDL
Simulation	Siemens QuestaSim 2025
Editor	VS Code

Part 2: UART Protocol, Architecture and RTL Design

5. UART Protocol

UART (Universal Asynchronous Receiver Transmitter) is an asynchronous serial communication protocol. In this project, communication follows the standard UART frame consisting of one start bit, eight data bits and one stop bit. Data is transmitted Least Significant Bit (LSB) first. Both transmitter and receiver operate using the same baud rate generated by the baud generator.

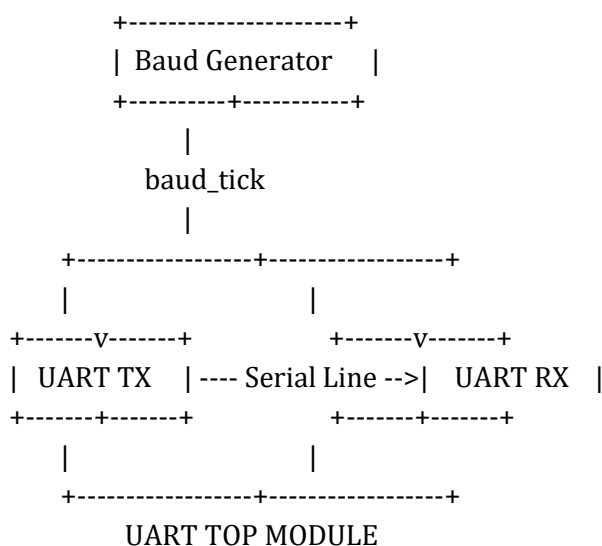
UART Frame Format



Transmission Process

- Line remains HIGH during idle.
- Transmission begins with START bit (0).
- Eight data bits are transmitted LSB first.
- STOP bit (1) completes the frame.
- Transmitter returns to IDLE.

6. System Architecture



The UART Top module integrates the baud generator, UART transmitter and UART receiver. The baud generator provides timing pulses to both transmitter and receiver. The transmitter serializes parallel data while the receiver reconstructs the received serial stream into parallel data.

7. RTL Module Documentation

7.1 baud_generator.v

Purpose: Generate a baud_tick pulse after BAUD_DIV clock cycles.

Inputs: clk, rst

Output: baud_tick

Working: A counter increments on every clock cycle. When the counter reaches BAUD_DIV-1, baud_tick is asserted for one clock cycle and the counter resets.

7.2 uart_tx.v

Purpose: Serially transmit 8-bit parallel data.

FSM States: IDLE, START, DATA and STOP.

Working: On tx_start, data is loaded into the shift register. On every baud_tick one bit is transmitted until all eight bits are sent followed by the stop bit.

7.3 uart_rx.v

Purpose: Receive UART serial data.

Working: Detect the start bit, sample incoming bits at each baud tick, reconstruct the byte and assert rx_done after successful reception.

7.4 uart_top.v

Purpose: Integrate all RTL modules into a complete UART subsystem.

Working: Connect baud generator, transmitter and receiver using common timing signals.

Part 3: Verification, Simulation and Design Analysis

8. Verification Methodology

Version 1 verification was performed using a directed Verilog testbench. The objective was to verify correct end-to-end UART communication by transmitting an 8-bit data value through the UART transmitter and confirming successful reception at the UART receiver.

8.1 Verification Environment

- Simulator: Siemens QuestaSim 2025 (EDA Playground)
- Language: Verilog HDL
- Design Under Test (DUT): uart_top
- Testbench: uart_top_tb.v
- Verification Style: Directed testing

8.2 Testbench Description

The testbench generates a system clock, applies an asynchronous reset, provides transmit data, asserts the tx_start signal, and monitors the UART receiver outputs. The simulation continues until one complete UART frame has been transmitted and received.

8.3 Test Sequence

1. Initialize all signals.
2. Apply reset.
3. Release reset after a few clock cycles.
4. Load 8-bit transmit data.
5. Assert tx_start.
6. Wait for UART transmission to complete.
7. Observe received data and completion signal.
8. Finish simulation.

9. Simulation Results

Functional simulation confirmed successful UART communication. The transmitter generated the expected UART frame consisting of a start bit, eight data bits (LSB first), and a stop bit. The receiver reconstructed the serial data correctly.

9.1 Waveform Analysis

- Reset initializes all RTL modules.

- baud_tick is generated periodically.
- tx remains HIGH during idle.
- Transmission begins when tx_start is asserted.
- Start bit is transmitted LOW.
- Eight data bits are shifted out LSB first.
- Stop bit returns tx to HIGH.
- Receiver samples incoming serial data.
- Received byte matches transmitted byte.
- Simulation completes successfully without errors.

9.2 Waveform Placeholder

Insert your captured waveform image (uart_top_waveform.png) in this section. Add a figure caption such as:

Figure 9.1: Functional simulation waveform of the integrated UART Version 1 design.

10. Design Decisions

- Separated functionality into independent RTL modules to improve readability and reuse.
- Implemented UART TX using a finite state machine for predictable control flow.
- Used a shift register for serial transmission of parallel data.
- Generated baud timing using a parameterized counter-based baud generator.
- Maintained synthesizable coding practices throughout the RTL implementation.
- Verified complete system functionality using an integrated top-level testbench.

Part 4: Conclusion, Limitations and References

11. Limitations of Version 1

Version 1 establishes a functional UART RTL implementation and verification environment. The following features are intentionally not included in this release:

- Parity bit generation and checking
- Configurable stop bits
- FIFO buffering
- Error detection (framing, parity, overrun)
- Interrupt support
- SystemVerilog assertions
- Functional coverage
- UVM-based verification

12. Future Enhancements

The following enhancements are planned for future versions of the project:

- RTL optimization and code refinement (Version 2)
- Improved finite state machine implementation
- SystemVerilog-based verification environment
- Assertion-Based Verification (ABV)
- Functional coverage collection
- Constrained-random verification
- Universal Verification Methodology (UVM)
- Regression testing
- Continuous integration using GitHub

13. Conclusion

Version 1 of the UART Design and Verification Project successfully implements a modular UART communication subsystem using synthesizable Verilog HDL. The design includes a baud rate generator, UART transmitter, UART receiver and top-level integration module. Functional verification was completed using a directed Verilog testbench and simulation in Siemens QuestaSim 2025 via EDA Playground. Waveform analysis confirmed successful serial transmission and reception, demonstrating correct operation of the integrated RTL design. The modular architecture developed in Version 1 provides a solid foundation for future RTL improvements and advanced verification using SystemVerilog and UVM.

14. References

- UART Specification (general reference).
- IEEE Std 1364 – Verilog Hardware Description Language.
- Siemens QuestaSim User Documentation.
- EDA Playground Documentation.
- Course notes and self-developed RTL implementation.

Appendix A – Project Directory Structure

```
UART_UVM_Project/
|
├── Version_1/
|   ├── RTL/
|   |   ├── baud_generator.v
|   |   ├── uart_tx.v
|   |   ├── uart_rx.v
|   |   └── uart_top.v
|   ├── TB/
|   |   └── uart_top_tb.v
|   ├── Waveforms/
|   |   └── uart_top_waveform.png
|   └── README.md
```

Document Status

Document Version : 1.0

Project Status : Version 1 Complete

RTL Status : Verified

Simulation : Passed

Next Milestone : Version 2 RTL Optimization